

MAKAR BADERKO

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ABOUT ME

Im a 17-year-old software engineer and AI researcher with an interdisciplinary background in mathematics, computer science, economics, and physics. My work focuses on machine learning and data-driven modeling.

SKILLS

ML & AI	PyTorch, Classic ML, Gradient Boosting, Data Mining, Time-series Modelling, Deep Learning, Computer Vision, LLM Fine-Tuning, Graph Neural Networks
Quantitative Economics	Mathematical Modelling, Statistical Experiment Design, Stochastic Optimisation
Economics	Econometric Analysis, Financial Modelling
Languages	Russian – Native, English – C1 (Cambridge certified), German – B2

RESEARCH

- AdLoCo: Adaptive batching significantly improves communications efficiency and convergence for Large Language Models
Nikolay Kutuzov, Makar Baderko*, Stepan Kulibaba*, Artem Dzhililov, Daniel Bobrov, Maxim Mashtaler, Alexander Gasnikov*
October 2025, ICOMP Proceedings
- Exploring Applications of State Space Models and Advanced Training Techniques in Sequential Recommendations: A Comparative Study on Efficiency and Performance
Mark Obozov, Makar Baderko*, Stepan Kulibaba*, Nikolay Kutuzov, Alexander Gasnikov*
October 2024, ICOMP Proceedings

ACADEMIC ACHIEVEMENTS

Computer Science and AI

- National Statistics and Data Analysis Olympiad – Gold(2022, 2024, 2025), Silver(2023)
Top-1 result (among my grade participants) in the theoretical section in 2022 and 2024
All-time national cumulative score record holder
This is the most significant olympiad in data analysis in Russia
- Kaggle Competitions Master (top-300 out of 200,000)
Gold Medal in "March Machine Learning Mania 2026" (top-7 out of 3,462 teams)
Silver Medal in "Deep Past Challenge 2026" (top-33 out of 2,673 teams)
Silver Medal in "AI Mathematical Olympiad – Progress Prize 3" (top-23 out of 4,138 teams)
- Autumn National Olympiad in AI (IAIO 2026 National Qualification) – 5th in the country
- National Technological Olympiad in Artificial Intelligence – Gold(2025), top-1 result (among my grade participants)
This was the most significant olympiad in AI in Russia at the time of participation
- AIJ Junior Contest – Winner (1st place), Computer Vision case (2023)
- Big Challenges Conference on Big Data and Machine Learning – Prize-winner(2024)
This is the most significant high school conference in the field of Computer Science in Russia
- Sirius.AI Conference – Laureate for the best scientific problem solution (2023)
- Northern Eurasia Olympiad in AI – second best result in the Classic Computer Vision task (2025)
- National Technological Initiative Talent nominee – Engineering (2025), Software Engineering (2025) and AI (2025)

Physics and Mathematics

- Atomic Industry Physics Olympiad (held by Rosatom) – top-12 in the country in 2024, Winner(2024), Prize-winner(2025, 2026)
- Winner and Prize-winner of 7 Physics Olympiads held by MIPT, MEPhI, and BMSTU (2024-2026)
- Winner and Prize-winner of 3 Mathematics Olympiads held by BMSTU (2024-2026)

Economics and Business

- WEC Continental Economics Cup – 2nd place in Europe in the Business Case Analysis section (scored 190/200) and a Bronze medal overall (2024)
- International Economics Olympiad (Open Track) – 95% in Finance and 98.9% in Economics (top 2 out of 500 in economics in the world)
- Moscow State University Championship in Economics – PwC Russia nomination awardee (2025)
- Winner and Prize-winner of 3 Economics Olympiads held by the Higher School of Economics (2023-2026)

Foreign languages

- The Highest Probe Olympiad in English Language (held by HSE) – Winner(2025)
- National Olympiad in German Language – Finalist (2026), top 0.2% in Russia

EDUCATION

Center for Teacher Excellence School

Physics Major, GPA 5.0/5.0

(2024 - Present)

- Best (top-1) school in Russia by the number of National Olympiad awardees with a highly-competitive admission process
- Enrolled in elective courses in Advanced Maths, Advanced Calculus and two different courses in olympiad-level Physics
- Awarded a merit-based tuition fee scholarship

"Vybor" (Choice) Private School

GPA 5.0/5.0

(2015 - 2024)

- Took electives in Statistics, Advanced Maths, Computer Science, Advanced Physics, English and German languages
- Full score (100%) on the middle school leaving qualification exams in Mathematics and Computer Science

Granted merit-based admission into the world's best (top-1) International Baccalaureate school (Letovo school).

Additional Studies

- Summer competitive programming training at MIPT
Learned dynamic programming, graph traversals, data structures and combinatorics
Placed 2nd in the final team-based algorithmic programming competition (ICPC-format)
- A year-long Data Structures and Algorithms course at Yandex, Russia's largest IT company
Learned core data structures, algorithm design techniques, complexity analysis, and problem-solving through programming assignments and contests.
- A year-long olympiad-level physics training course at the Moscow Physics Team
Solved various National Olympiad & IPhO level problems in theoretical and experimental physics
Qualified for IEPHO

WORK EXPERIENCE

Laboratory of Mathematical Methods of Optimisation, MIPT

Machine Learning Research Intern

(June 2025 - August 2025)

- Co-developed AdLoCo, a novel distributed optimisation framework that significantly improves communication efficiency and convergence for LLMs
- Contributed to the mathematical proof of communication complexity bounds for non-convex stochastic optimisation

Educational Scientific Center Sirius

Machine Learning Research Intern

(July 2024 - October 2024)

- Applied Hydra bidirectional and Mamba/Mamba2 architectures to the problem of sequential recommendation, outperforming SOTA models in latency & NDCG
- Managed model fine-tuning on a GPU cluster, utilizing complex custom-developed schedulers and adaptive batching techniques

Biometriclabs LLC

Software Engineering Intern (Distributed Systems)

(July 2023 - August 2023)

- Created a high-performance codebase to connect the mathematical models written in C with the existing UI and the database of reports from sensors

Software Engineering Intern (Embedded Systems)

(June 2021 - August 2021)

- Built prototypes of innovative products with the Raspberry Pi, collected human heart rate data and converted it into the required digital format

Granted direct admission to internships in ML at largest Russian banks (Sber and Tbank) for winning top corporate ML competitions (interview waived).

PROJECTS

Computer Science and AI

- Machine learning model that uses publicly available blockchain network data to prevent a transaction with a high risk of money laundering (GNNs and more)
- Amur tiger detection, segmentation and reidentification software for remaining population size estimation
- Unwanted plants dataset collection, detection and segmentation software for farmers
- Computer Vision system to detect people who have fallen on the railroad (for the Russian Railways)

- Retrieval-augmented generation (RAG) system that enhances large language model reasoning on mathematical problems for the Unified State Exam commission
- Parcel weight and size estimation by photo and description using large language models and computer vision techniques for Avito, the largest classified ads platform in Russia
- NCAA Basketball matches outcome estimation using past match performance, bookmaker odds and machine learning (Kaggle Gold Medal)
- Translation of the business records of ancient Assyrian merchants from Akkadian into English using NLP techniques (Kaggle Silver Medal)
- Applied LLM-based reasoning to solve advanced math problems and develop solution strategies for the AIMO3 AI mathematical reasoning challenge (Kaggle Silver Medal)
- Fine-tuning Large Language Models (LLMs) to bypass state-of-the-art results on Q&A benchmarks with reasoning using advanced training techniques (QLoRA, DPO, etc.)

Economics and Finance

- Company reviews reliability evaluation for credit scoring using advanced mathematical modelling techniques using data about 15,000+ companies
- Governmental investment effectiveness evaluation and prediction based on overall economy performance using data about 200,000 companies
- Private investment regularity evaluation and prediction for brokerage recommender systems using data about 5000 clients
- Statistical call-centre customer satisfaction analysis for a major Russian bank (explored patterns in waiting times)
- Modelling the relationship between infrastructure development and real estate prices in New Moscow for a SaaS analytics company
- Statistical analysis of the factors influencing students evaluations of teachers academic qualities at the HSE University